



Advanced Linear Algebra: Foundations to Frontiers

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3.2.3 Classical Gram-Schmidt algorithm ¶

03.2.3 CGS Algorithm in FLAME notation

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Remark 3.2.3.1. If the FLAME notation used in this unit is not intuitively obvious, you may want to review some of the materials in Weeks 3-5 of Linear Algebra: Foundations to Frontiers (<http://www.ulaff.net>).

An alternative for motivating that algorithm is as follows:

- Consider $A = QR$.
- Partition A , Q , and R to yield

$$\left(\begin{array}{c|cc} A_0 & a_1 & A_2 \end{array} \right) = \left(\begin{array}{c|cc} Q_0 & q_1 & Q_2 \end{array} \right) \left(\begin{array}{c|cc} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ \hline 0 & 0 & R_{22} \end{array} \right).$$

- Assume that Q_0 and R_{00} have already been computed.
- Since corresponding columns of both sides must be equal, we find that

$$a_1 = Q_0 r_{01} + q_1 \rho_{11}. \quad (3.2.1)$$

Also, $Q_0^H Q_0 = I$ and $Q_0^H q_1 = 0$, since the columns of Q are mutually orthonormal.

- Hence

$$Q_0^H a_1 = Q_0^H Q_0 r_{01} + Q_0^H q_1 \rho_{11} = r_{01}.$$

- This shows how r_{01} can be computed from Q_0 and a_1 , which are already known:

$$r_{01} := Q_0^H a_1.$$

- Next,

$$a_1^\perp := a_1 - Q_0 r_{01}$$

is computed from (3.2.1). This is the component of a_1 that is perpendicular (orthogonal) to the columns of Q_0 . We know it is nonzero since the columns of A are linearly independent.

- Since $\rho_{11} q_1 = a_1^\perp$ and we know that q_1 has unit length, we now compute

$$\rho_{11} := \|a_1^\perp\|_2$$

and

$$q_1 := a_1^\perp / \rho_{11},$$

These insights are summarized in the algorithm in Figure 3.2.3.2.

$[Q, R] = \text{CGS-QR}(A)$
$A \rightarrow \left(A_L \mid A_R \right), Q \rightarrow \left(Q_L \mid Q_R \right), R \rightarrow \left(\begin{array}{c c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right)$ A_L and Q_L has 0 columns and R_{TL} is 0×0 while $n(A_L) < n(A)$ $\left(A_L \mid A_R \right) \rightarrow \left(A_0 \mid a_1 \mid A_2 \right), \left(Q_L \mid Q_R \right) \rightarrow \left(Q_0 \mid q_1 \mid Q_2 \right),$ $\left(\begin{array}{c c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c cc} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ 0 & 0 & R_{22} \end{array} \right)$ $r_{01} := Q_0^H a_1$ $a_1^\perp := a_1 - Q_0 r_{01}$ $\rho_{11} := \ a_1^\perp\ _2$ $q_1 := a_1^\perp / \rho_{11}$ $\left(A_L \mid A_R \right) \leftarrow \left(A_0 \mid a_1 \mid A_2 \right), \left(Q_L \mid Q_R \right) \leftarrow \left(Q_0 \mid q_1 \mid Q_2 \right),$ $\left(\begin{array}{c c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c cc} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ 0 & 0 & R_{22} \end{array} \right)$ endwhile

Figure 3.2.3.2. (Classical) Gram-Schmidt (CGS) algorithm for computing the QR factorization of a matrix A .

Having presented the algorithm in FLAME notation, we can provide a formal proof of [Theorem 3.2.2.1](#).

[Proof of Theorem 3.2.2.1.](#)

Informal proof: The process described earlier in this unit constructs the QR decomposition. The computation of $\rho_{j,j}$ is unique if it is restricted to be a real and positive number. This then prescribes all other results along the way.

Formal proof:

(By induction). Note that $n \leq m$ since A has linearly independent columns.

- Base case: $n = 1$. In this case $A = \left(A_0 \mid a_1 \right)$, where A_0 has no columns. Since A has linearly independent columns, $a_1 \neq 0$. Then

$$A = (a_1) = (q_1)(\rho_{11}),$$

where $\rho_{11} = \|a_1\|_2$ and $q_1 = a_1/\rho_{11}$, so that $Q = (q_1)$ and $R = (\rho_{11})$.

- Inductive step: Assume that the result is true for all A_0 with k linearly independent columns. We will show it is true for A with $k + 1$ linearly independent columns.

Let $A \in \mathbb{C}^{m \times (k+1)}$. Partition $A \rightarrow \left(A_0 \mid a_1 \right)$.

By the induction hypothesis, there exist Q_0 and R_{00} such that $Q_0^H Q_0 = I$, R_{00} is upper triangular with nonzero diagonal entries and $A_0 = Q_0 R_{00}$. Also, by induction hypothesis, if the elements on the diagonal of R_{00} are chosen to be positive, then the factorization $A_0 = Q_0 R_{00}$ is unique.

We are looking for

$$\left(\tilde{Q}_0 \mid q_1 \right) \text{ and } \left(\begin{array}{c|c} \tilde{R}_{00} & r_{01} \\ \hline 0 & \rho_{11} \end{array} \right)$$

so that

$$\left(A_0 \mid a_1 \right) = \left(\tilde{Q}_0 \mid q_1 \right) \left(\begin{array}{c|c} \tilde{R}_{00} & r_{01} \\ \hline 0 & \rho_{11} \end{array} \right).$$

This means that

$$\circ A_0 = \tilde{Q}_0 \tilde{R}_{00},$$

We choose $\tilde{Q}_0 = Q_0$ and $\tilde{R}_{00} = R_{00}$. If we insist that the elements on the diagonal be positive, this choice is unique. Otherwise, it is a choice that allows us to prove existence.

- $a_1 = Q_0 r_{01} + \rho_{11} q_1$ which is the unique choice if we insist on positive elements on the diagonal.

$a_1 = Q_0 r_{01} + \rho_{11} q_1$. Multiplying both sides by Q_0^H we find that r_{01} must equal $Q_0^H a_1$ (and is uniquely determined by this if we insist on positive elements on the diagonal).

- Letting $a_1^\perp = a_1 - Q_0 r_{01}$ (which equals the component of a_1 orthogonal to $\mathcal{C}(Q_0)$), we find that $\rho_{11} q_1 = a_1^\perp$. Since q_1 has unit length, we can choose $\rho_{11} = \|a_1^\perp\|_2$. If we insist on positive elements on the diagonal, then this choice is unique.
- Finally, we let $q_1 = a_1^\perp / \rho_{11}$.

- By the Principle of Mathematical Induction the result holds for all matrices $A \in \mathbb{C}^{m \times n}$ with $m \geq n$.

Homework 3.2.3.1. Implement the algorithm given in [Figure 3.2.3.2](#) as

function [Q, R] = CGS_QR(A)

by completing the code in [Assignments/Week03/matlab/CGS_QR.m](#). Input is an $m \times n$ matrix A . Output is the matrix Q and the upper triangular matrix R . You may want to use [Assignments/Week03/matlab/test_CGS_QR.m](#) to check your implementation.

▼ [Solution](#)

See [Assignments/Week03/answers/CGS_QR.m](#).
(Assignments/Week03/answers/CGS_QR.m)